



The Wrong Turn

Ontario Math C3.2 — read, find, and fix code

Name: _____

Date: _____

★ I can fix a turtle shape loop with the wrong turn angle.

Wrong turn, open shape

In a turtle shape, each shape needs the right turn: square 90, triangle 120, pentagon 72, hexagon 60.

If the turn is wrong, the shape will not close. The range can be right and it still looks wrong — check the turn.

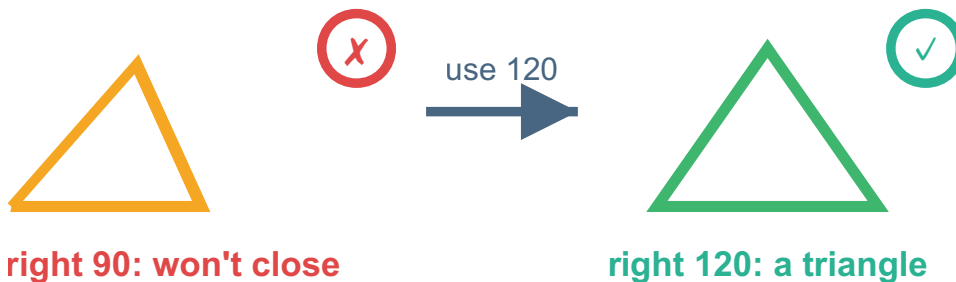
This should draw a TRIANGLE — but it's buggy

```
for i in range(3):  
    bit.forward(50)  
    bit.right(90)
```

PYTHON

OUTPUT

The corners are wrong — the triangle does not close.



right 90 won't make a triangle; right 120 closes it.

1 Find the bug

The triangle code uses right(90). A triangle needs right 120. What should right() be? _____

2 Fix the hexagon

This should draw a hexagon (6 sides): for i in range(6): bit.forward(40); bit.right(90). A hexagon needs right 60. What is wrong, and what should the turn be?

3 Write the fix

Bit should draw a pentagon (5 sides). The turn for a pentagon is 72. Write it: bit.right(_____)

4 Challenge

A SQUARE loop has right(60) instead of right(90). Will the square close? Why not? What should the turn be?

Big idea

Each shape has its own turn. If the shape does not close, the turn is probably wrong — match the turn to the shape (square 90, triangle 120, ...).